Final Project Report

AI vs Human Text Detection

Clara Obeng-Akrofi

1. Introduction

Generative AI tools like ChatGPT, Bard, and Claude have become central to research, learning, information seeking, and writing. As large language models grow more sophisticated and widely accessible, distinguishing between human-written and machine-generated text has become a critical challenge. This distinction matters greatly in domains where authenticity is important for trust, accountability, and integrity.

Educational institutions worry about academic dishonesty as students increasingly use AI for assignments. News organizations face the spread of synthetic misinformation at unprecedented speed and scale. Scientific publishers must safeguard research integrity against fabricated submissions. Even legal and medical sectors, where accuracy and authorial responsibility are essential, now confront highly convincing AI-generated documents.

The task of detecting AI-generated text is inherently complex. As detection methods improve, models also evolve and develop techniques to evade detection. A large number of research papers have focused on identifying AI-generated writing through signals such as em dashes, vague phrasing, or overly sophisticated language. However, far less attention has been given to detecting *human* written text. This project aims to address that gap by exploring how well text-mining methods can classify AI-generated and human-written text, which types of features are most informative, and whether certain writing styles are easier or harder to distinguish.

2. Literature Review

Existing research on AI-generated text detection has grown rapidly in response to the widespread use of large language models. Several scholars have attempted to understand what makes AI text detectable and which linguistic or structural markers can reliably distinguish human writing from machine-generated text. This study builds on that foundation by examining how well a classifier can learn these distinctions and by identifying the specific features that most strongly signal human written text.

Ippolito et al. (2020) present one of the earliest systematic examinations of AI-generated language detection. The paper found that machine-generated texts which successfully mimic human written text are also the most difficult for computational models to classify correctly. More recently, Mitchell et al. (2023) developed DetectGPT, which detects AI text by analyzing the curvature of the probability distribution of the generated text. While the model's performance was impressive, a very notable weakness is that the model requires excessive computing power.

Hayawi et al. (2023) examined how genre affects the accuracy of AI-text detectors and found that certain writing domains, such as narrative and opinion pieces, are significantly more challenging to classify. This is particularly relevant to our study, as our dataset includes texts from multiple topical categories, including politics and technology. Finally, Jakesch et al. (2023) found that humans often rely on misleading indicators when identifying AI text. For example, readers may assume that certain tones or details indicate human authorship, even though AI can produce such content as well.

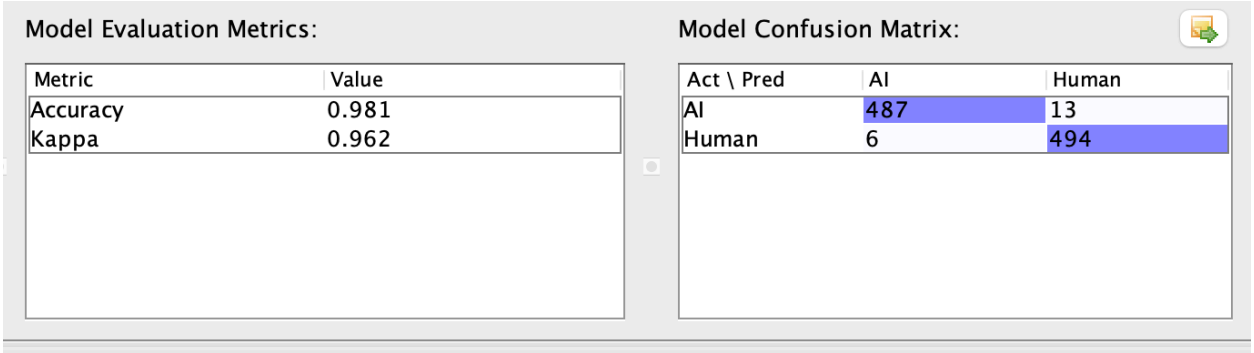
These studies provide the conceptual foundation for this project. This project contributes to this ongoing conversation by evaluating which features most reliably detect human written text.

3. Dataset

The dataset has approximately 487,000 text samples, each labeled as either human-written (0) or AI-generated (1). Out of these, about 305,000 are human-written and 181,000 are AI-generated. For the purposes of this analysis, a balanced stratified subset of 1,000 total samples was used for the test set and another of 1000 total samples for the training set. The data was split into a training set and a test set, allowing the classifier to learn patterns from the training portion and then evaluate its performance.

The texts varied widely in topic, including themes such as politics, the environment, transportation, personal reflections, and general social issues. This diversity was valuable for model training because it exposed the classifier to a broad range of linguistic styles and rhetorical structures, which is helpful for ensuring the classifier generalizes well in other test instances.

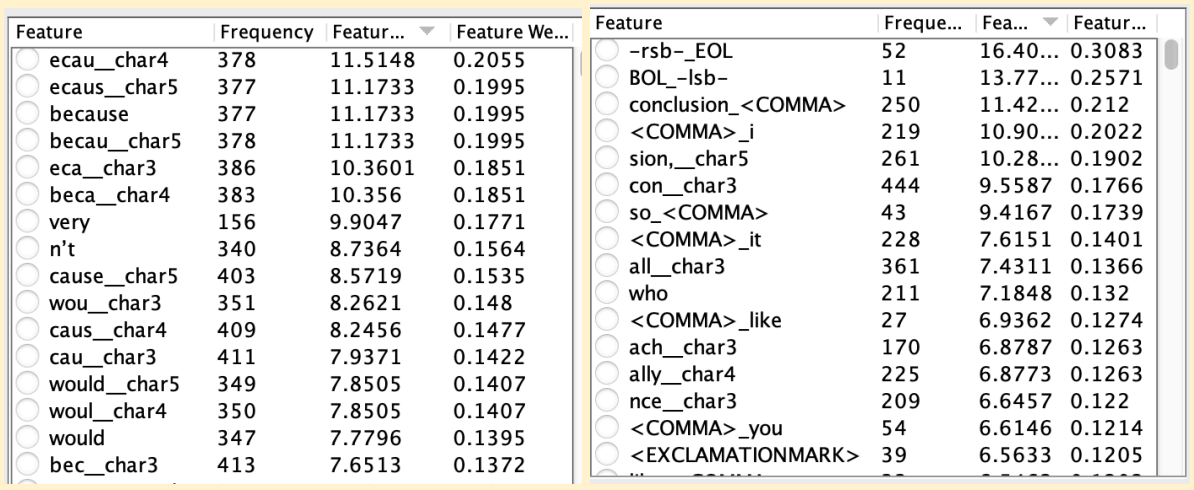
4. Approaches & Results



The initial model performed incredibly well, reaching 81% accuracy and 79% Cohen's Kappa using unigrams and bigrams with logistic regression. So, the main work of our project was error analysis to improve the model. The model consistently mistook two main things: mislabeling a document as human when it contained a quote from a human and not being able to attribute misspellings to any group. To combat this, we ran a python script to clean the texts, removing quotation marks, blocking quotes, and long embedded citations that inflated a document's "human" tendencies. To address misspelling we implemented rare-feature threshold and n-grams that would be able to tokenize misspellings and with logistic regression, these features became "highly predictive". However, this created a sizable amount of noise. To offset this, we set a minimum frequency cutoff to filter tokens that appeared less than two times in the dataset. This step removed almost all human-generated typos. We had initially thought typos would be a strong indicator of humans, but we were not able to capture the wide array of possible misspellings. The ones our model did catch led to overfitting.

Our final model relied on tokenization with logistic regression, specifically subword patterns, punctuation structures, and clause markers in order to find the stylistic difference between AI generated and human texts. This resulted in an accuracy of 98.1% and a Cohen's Kappa of 96.2%.

5. Best Features for Human Classification



Feature	Frequency	Featur...	Feature We...
ecau__char4	378	11.5148	0.2055
ecaus__char5	377	11.1733	0.1995
because	377	11.1733	0.1995
becau__char5	378	11.1733	0.1995
eca__char3	386	10.3601	0.1851
beca__char4	383	10.356	0.1851
very	156	9.9047	0.1771
n't	340	8.7364	0.1564
cause__char5	403	8.5719	0.1535
wou__char3	351	8.2621	0.148
caus__char4	409	8.2456	0.1477
cau__char3	411	7.9371	0.1422
would__char5	349	7.8505	0.1407
woul__char4	350	7.8505	0.1407
would	347	7.7796	0.1395
bec__char3	413	7.6513	0.1372

Feature	Freque...	Fea...	Featur...
-rsb-_EOL	52	16.40...	0.3083
BOL_-lsb-	11	13.77...	0.2571
conclusion_<COMMA>	250	11.42...	0.212
<COMMA>_i	219	10.90...	0.2022
sion,__char5	261	10.28...	0.1902
con__char3	444	9.5587	0.1766
so_<COMMA>	43	9.4167	0.1739
<COMMA>_it	228	7.6151	0.1401
all__char3	361	7.4311	0.1366
who	211	7.1848	0.132
<COMMA>_like	27	6.9362	0.1274
ach__char3	170	6.8787	0.1263
ally__char4	225	6.8773	0.1263
nce__char3	209	6.6457	0.122
<COMMA>_you	54	6.6146	0.1214
<EXCLAMATIONMARK>	39	6.5633	0.1205

The N-gram subword pattern extraction found several highly predictive words and stylistic choices that differentiated human and AI text. A quick feature to speak about was "-rsb-_EOL", meaning right square bracket and end of line. AI writing often has bracketed citations, inline references, and generated headers/list markers. AI writing tended to be more professional, using the suffixes "-ally" and "-nce" 435 times compared to humans using them only 155 times. These suffixes are nominalizations that appear in formal vocabulary: "critically", "systematically", "importance", "reference", and "significance". AI generated academic, polished, and generalized text. This is because they are trained on heavily edited and high quality texting, leading to an often instructional register. This results in no emotional nuance, personal voice, or casual reasoning.

AI also continually used discourse markers like "In conclusion" and "So," often to begin a more argumentative or informative statements. These are overt signs that the text is transitioning from one logical unit to another. This is not human relational thinking. It is logical separation, so the transition must be announced. The formulaic writing is optimized for clarity and coherence, so structure has a specific default format. The final major predictive feature is "so,". Interestingly, this was used more by humans than AI, but logistic regression takes into account conditional probability, which we will discuss later.

For human text, the model found casualness, emotion, and causality to be the most predictive of human texts. Humans tend to overuse words like "so", "because", "would", "were", and contractions like "n't". At first glance, these words may seem to indicate simply informality, but digging deeper we can find a reflection of the way humans communicate thoughts and emotions. "Because" is a deeply relational word. It has a

cause and effect pattern, explaining how or why something happened. This is the general human way of thinking, where your emotions and thoughts are determined by a myriad of things and how they interact with each other. AI treats things independently of each other. "This thing happened. *Therefore*, this other thing." These two things are independently occurring rather than have a relationship. AI organizes occurrences independently in order to avoid information overlap, and we can see that reflected in the model. Humans blend things together messily, which is why we see much more conversational text be predictive towards humans. "Because" has 386 human instances in the dataset compared to only 90 by AI.

A very similar thing is happening with "would", a word that conveys uncertainty, possibility, and hypothetical thinking: "What *would* you do?", "That *would* mean…", "It *would* mean a lot..". These statements have emotional nuance, as human language reflects human thought. AI speaks with authority and certainty, so it does not need words that convey these sentiments. The model was able to pick up on these subtleties, and "would" became a human marker. You may think that this is based on frequency, since it appears in human text 354 times compared to 137 times in AI, but more realistically the model was able to determine humans use subjective positioning to display emotion.

All together, these features reveal a clear stylistic difference between the two classes. AI organizes information and maintains certainty in text, while humans often display their own emotions and relations. AI has been trained for an optimized clear and coherent output that relies on professionalism, discourse markers, and structural transitions in an attempt to organize logic. Humans tend to reveal themselves in their writing. Text is based on and mirrors the writer's thought process and emotions. All of this shows that there is a fundamental difference in cognition between AI and humans. Our classifier was able to identify beyond words and capture ways of thinking.

6. Evaluation Metrics

113 AI	AI	0.9999281440607364	7.185593926362088...	Education in the key to...
114 AI	Human	0.483451847477633...	0.5165481525223667	Education is a lifelong ...
115 AI	AI	0.9992121356267029	7.878643732971424...	Careers play a signific...
116 Human	Human	0.005336785728692...	0.9946632142713071	All students should par...
117 AI	Human	0.385487126564847...	0.6145128734351522	IE today's society, it is ...

To assess the performance of our classifier, we used several evaluation metrics to prove how well the model generalized from the training data. Accuracy is generally inefficient for this task since there are only two categories, even if the sets are balanced. Instead, we prioritized Cohen's Kappa, which received a 96.2% rating. Kappa is important as it

measures agreement between predicted and true labels while accounting for the expected agreement level by chance. This works well for binary classification.

Besides Kappa, we strongly took into account label distributions after the model ran (pictured above). The blue on the third column indicates the label was correctly predicted, while orange represents an incorrect prediction. The numerical values are the model's AI-to-Human probability scores for each document. Examining these scores allowed us to evaluate not just whether the classifier was right, but how confident or reasonable its decisions were. In earlier models, score distributions were polarized. Documents received scores like 0.9999 or 0.0001 regardless of whether they were actually labeled correctly. This means the classifier had latched on to a small number of dominant features, leading it to make overconfident predictions.

Now looking at the final model, misclassified documents tended to fall in the middle of the distribution. This demonstrated a well-calibrated regression model that is able to make predictions like a human: instead of being wholly wrong, it is simply uncertain. This suggests that the remaining mislabeled documents were not caused by incorrect features. They were most likely caused by genuine ambiguity in the data.

The label distribution was able to act as a calibration check throughout the development of the model, proving our model was making nuanced decisions. It also lends credence to the high Kappa score. The model was not memorizing or overfitting. It was finding meaningful structures and behaviour between AI generated and Human text.

7. Post Error Analysis

While the model performed extremely well, we did not reach 100% accuracy. So in this section, we will discuss a couple documents that were mislabeled by the final model.

"Education is a life-long process, and I strongly agree with this statement. Education is not just about learning facts and figures, it is about learning how to think and how to apply knowledge to everyday life. It is a process that should never end, as it helps us to grow and develop as individuals. First, education helps us to develop our critical thinking skills. It teaches us how to analyze and evaluate information, how to form our own opinions, and how to make informed decisions."

This document is AI generated, but was labeled Human 51.6 to 48.3. From analysis of how the model works, we can claim that this text is fairly middle of the road with no strong indicators of being AI or Human either way. You may assume human as it contains

markers like opinion framing, and structured argument transitions, but the purpose and importance of this project is to prove that AI generated text has become advanced enough to mimic the styling of humans enough to where simple intuition cannot determine the correct labeling.

8. Discussion

Our best classifier, logistic regression, achieved an accuracy of 0.98 on the test set, which is impressive given the overlap between human and AI writing styles. This suggests that even as AI models grow more sophisticated, they still leave behind detectable stylistic traces.

One of the surprising findings is that the strongest predictors of *human* writing were not grammatical structures or emotional tones, but very ordinary words such as “because,” “getting,” “being,” and “says”. These words appear often in natural human communication but tend to be used less by AI models.

At the same time, the error cases reveal the limitations of detection models. Human-written texts that were very polished or overly formal were often misclassified as AI-generated. Likewise, AI texts that mimicked conversational tone or introduced minor inconsistencies were sometimes labeled as human. These mistakes highlight a larger challenge: the boundary between human and AI writing continues to blur, especially when AI tools are prompted to write more casually or more imperfectly. As a result, any classifier will face difficulty once AI-generated text starts to intentionally resemble human linguistic patterns.

Another peculiar instance found by our model was the conditional context of the word “so.” Specifically, “so,” being highly predictive of AI text, while “so” was predictive of human text. This seems contradictory, but they were two of the most predictive features even though humans used “so” far more frequently. Logistic regression was one of the most important parts of our classifier. It was able to calculate how the presence of a feature shifts the probability towards a class while accounting for all other features present in the document. The comma after “so,” changes the meaning of the word and begins lending itself to the structural transitions AI notoriously uses: “So, to summarize” is an instance in the dataset. Humans, on the other hand, tended to use “so” as an informal connection: “So I thought”, “So then”, “so tired”. This remains on theme with the human proclivity to write relationally. More than anything, this example lends credence to the strength of logistic regression as a model. It was able to identity patterns rather than

isolated words. The conditional relationship of features is able to reveal habits of writers. Weighing features contextually is what lets our model succeed. This project was on the subtle differences between texts that could only be unveiled by discovering the reasoning behind them.

Overall, the findings suggest two things. First, detecting human text requires focusing on the features that reflect human variability rather than solely trying to identify the "AI fingerprint." Second, even simple features can be powerful when combined with a well-designed dataset and thoughtful preprocessing. However, as AI systems evolve, detection will need to continue adapting as well.

9. Conclusion

This project set out to explore what distinguishes human-written text from AI-generated text and how well a machine learning classifier can detect that difference. By using a balanced dataset of 2,000 samples, 1,000 for training and 1,000 for testing, and training a Logistic Regression classifier, we found that the model was able to classify text with a strong accuracy of 0.98. This shows that despite the rapid growth of large language models, meaningful differences between human and AI writing still exist and can be identified using text-mining techniques.

The best predictors of human writing were simple, everyday words and features that captured casualness in sentence structure and punctuation. Adjustments made during error analysis, such as removing quotes and adding line-length and punctuation features, played a big role in improving the classifier’s performance.

However, the project also exposed important limitations. AI-generated text is adapting at a fast pace, and is capable of mimicking human imperfections, which makes classification more challenging. Human writing can also appear machine-like when it is overly formal or clean. These overlaps show that detection cannot rely on a single set of stable features. Instead, it must evolve as writing tools evolve.

Reference

Ippolito, D., Krishna, S., Liu, C., & Callison-Burch, C. (2020). *Automatic detection of generated text is easiest when humans are fooled.* In *Proceedings of the 58th Annual Meeting of the*

Association for Computational Linguistics (pp. 1673–1684). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.164

Gehrmann, S., Strobelt, H., & Rush, A. M. (2019). *GLTR: Statistical detection and visualization of generated text.* In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: System Demonstrations* (pp. 111–116). Association for Computational Linguistics. https://doi.org/10.18653/v1/P19-3019

Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2023). *DetectGPT: Zero-shot machine-generated text detection using probability curvature.* arXiv. https://doi.org/10.48550/arXiv.2301.11305

Hayawi, K., Huang, H., & Aljahdali, H. (2023). *The imitation game: Detecting human and AI-generated texts in the era of ChatGPT and BARD.* https://doi.org/10.48550/arXiv.

Popkov AA, Barrett TS. AI vs academia: Experimental study on AI text detectors' accuracy in behavioral health academic writing. Account Res. 2025 Oct;32(7):1072-1088. doi: 10.1080/08989621.2024.2331757. Epub 2024 Mar 22. PMID: 38516933.

Jakesch, M., Darwish, K., & Hancock, J. T. (2023). *Human heuristics for AI-generated language are flawed.* Proceedings of the National Academy of Sciences. https://doi.org/10.1073/pnas.